

Basic Principles of Medicine 1

Module: Foundations to Medicine

DLA: Population Genetics:
Factors that cause deviations from Hardy-Weinberg Equilibrium

Please Watch After Lecture on Population Genetics

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Objectives

SOM.MKIII.BPM1.1.FTM.3.GNET.0082	Describe genetic drift
SOM.MKIII.BPM1.1.FTM.3.GNET.0083	Discuss founder effect
SOM.MKIII.BPM1.1.FTM.3.GNET.0084	Explain the meaning of heterozygote advantage
SOM.MKIII.BPM1.1.FTM.3.GNET.0085	Discuss how new mutation keeps incident rates stable if affected individuals have a lower genetic fitness
SOM.MKIII.BPM1.1.FTM.3.GNET.0086	Analyze how alleles segregate in populations and how to relate allele frequency in populations to genotype frequency of the people in the population
SOM.MKIII.BPM1.1.FTM.3.GNET.0087	Apply the Hardy Weinberg equation to clinical scenarios to allow estimation of carrier risk, recurrence risk, and frequency of disease alleles in a population for AR, AD, and X-linked genes.
SOM.MKIII.BPM1.1.FTM.3.GNET.0088	Estimate a carrier frequency from the Hardy Weinberg equation and population incidence for one individual to estimate recurrence risk when that person mates with an individual who has familial carrier risk.
SOM.MKIII.BPM1.1.FTM.3.GNET.0089	Use Hardy-Weinberg equation to demonstrate why the expected frequency of rare autosomal dominant alleles are expected to be extremely infrequent.

Population Genetics: Genotypes & Phenotypes

- Quantitative study: Distribution of genetic variation in the population
- Mathematical description of how the frequencies of genes and genotypes are maintained or changed
- Genetic factors
 - Mutation
 - Reproduction → fitness to reproduce
- Environmental and Societal factors
 - Selection
 - Migration
- Population genetics provides knowledge about different disease genes in different populations and assist with clinical diagnosis and genetic counseling

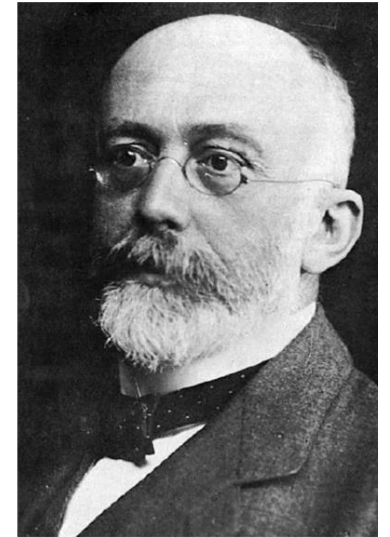
Assumptions of Hardy-Weinberg Equilibrium 1908

from Lecture

- Population is large
- Random mating between individuals
 - No preference for genotypes
- No mutations
 - No new alleles are generated
- No natural selection
 - All alleles have equal fitness
- No gene flow
 - No migration of individuals into the population



Mendel 1822-1884: work re-discovered in 1900



Hardy 1877-1947 & Weinberg 1862-1937

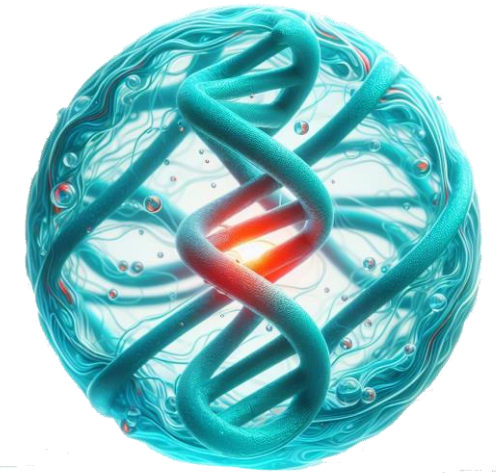
Factors Causing Genetic Variation in Populations

- New mutation
- Natural selection
 - Heterozygote advantage
- Non-random mating, inbreeding
- Gene flow
 - Emigration or immigration
- Small population sizes: Bottleneck
 - Genetic drift
 - Founder effect



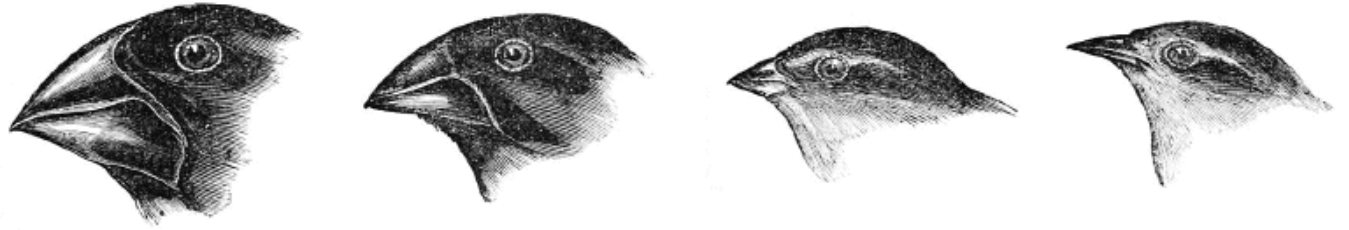
Leads to Departure from the Hardy-Weinberg Equilibrium

New Mutation

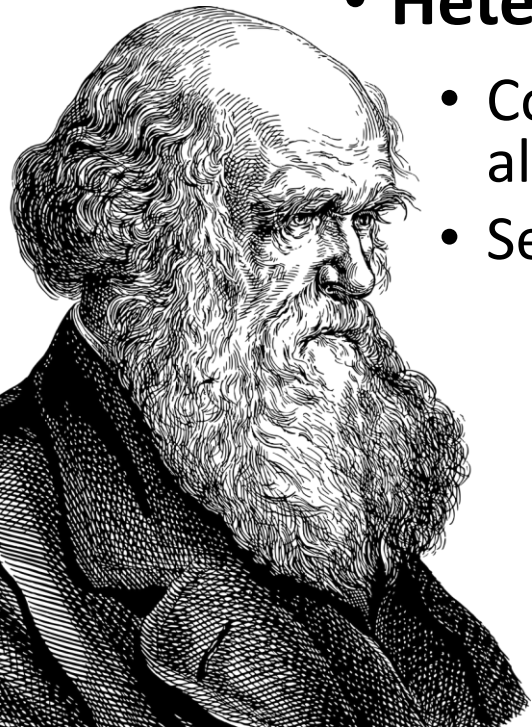


- Mutations arise in either a somatic or germ cell line
- **Harmful mutations** will be lost if they reduce reproductive fitness or cause early death
- **Harmful mutations** that arise from a mutational “hot spot” can become fixed in a population even though affected persons have reduced fitness and do not pass on the mutated allele (e.g. DMD)
- **Neutral mutations** come to equilibrium quickly in a large population
 - If environment changes, the neutral mutation may become beneficial
- **Beneficial mutations** may influence natural selection and may increase in allele frequency each generation

Natural Selection



- Selection typically eliminates deleterious variation from populations
- Mutations that are advantageous will be retained:
 - Change in allele frequencies and lead to change in genotype frequencies
- **Heterozygote Advantage:**
 - Consider Sickle Cell Anemia (SC), Cystic Fibrosis (CF), Tay-Sachs disease are all autosomal recessive disorders
 - Severe genetic diseases, yet persists at high frequency in the population
 - SC and CF patients now have increased lifespan due to medical advances, but fertility is affected for both conditions
 - Tay-Sachs fatal 2-5 years

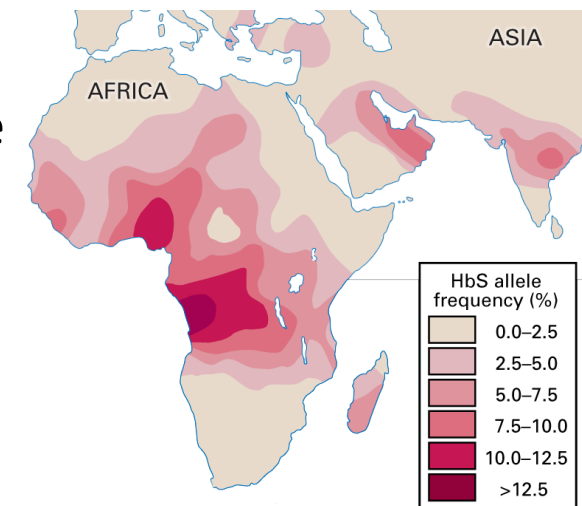
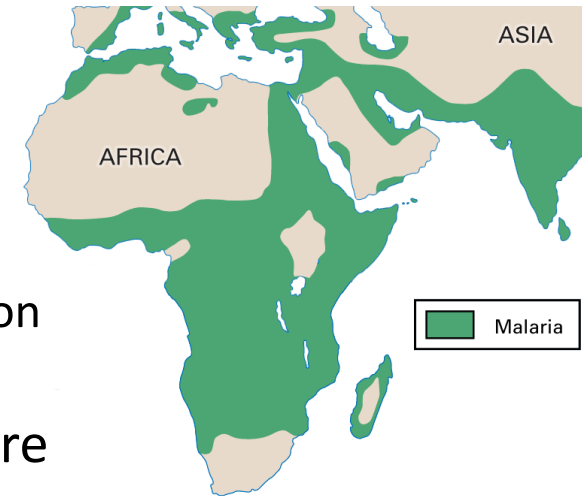


Darwin 1809-1882

SOM.MK.III.BPM1.1.FTM.3.GNET.0086

Natural Selection: Heterozygote Advantage Theory

- Severe autosomal recessive (aa) disorders that persist with low to no reproductive fitness
- Balance between selection and mutation?
- Possible that heterozygotes (Aa) have an advantage
 - Mathematical equations to calculate which I will spare you
 - Essentially heterozygote frequency higher than could be maintained by mutation because homozygous affected have low to no reproductive fitness
 - **SC** = High carrier frequency in people with ancestry from Africa and where malaria is endemic
 - Heterozygote advantage to withstand malaria infection as shown in maps→
 - **CF** = High carrier frequency in people with ancestry from eastern Europe that withstood many cholera and typhus pandemics
 - Heterozygote advantage to withstand severe diarrhea (cholera and typhus)
 - **TS** = High carrier frequency in people with Ashkenazi Jewish Eastern European ancestry due to segregation, crowded urban areas and lack of immigration.
 - Heterozygote advantage may give partial resistance to tuberculosis



Non-random Mating

- Persons related genetically are termed consanguineous and if they mate, progeny are said to be inbred
- Inbreeding increases the frequency of **homozygous genotypes** and decreases the frequency of heterozygous genotypes in the population
- Increased risk depends on the population frequency of the disease allele and the degree of relationship between parents
- E.g. Incidence of AR disorder is 1/40,000 in the population, first cousin parents
 - Calculate $p \times q \times F$, then add the Incidence

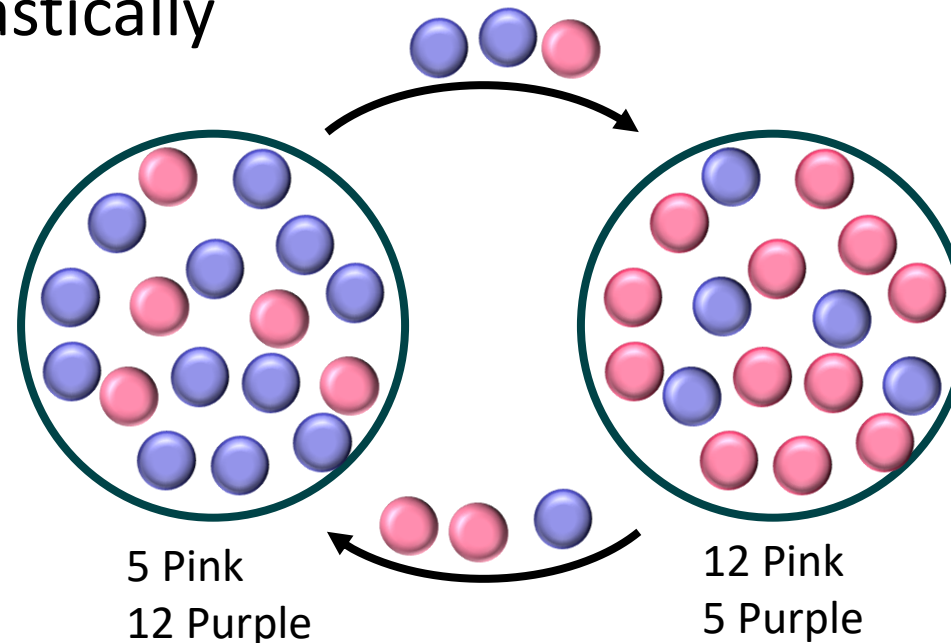
$$1/40,000 + (199/200)(1/200)(1/16) = 1/2977$$

Appreciate the increase in risk, and not the math

Mating Type	Proportion of Shared Alleles	Inbreeding coefficient F
Parent-offspring	$\frac{1}{2}$	$\frac{1}{4}$
Full siblings	$\frac{1}{2}$	$\frac{1}{4}$
Half siblings	$\frac{1}{4}$	$1/8$
Uncle-niece/Aunt-nephew	$\frac{1}{4}$	$1/8$
First cousins	$1/8$	$1/16$
Double first cousins	$\frac{1}{4}$	$1/8$
Half first cousins	$1/16$	$1/32$
First cousins once removed	$1/16$	$1/32$
Second cousins	$1/32$	$1/64$
Second cousins once removed	$1/64$	$1/128$
Third cousins	$1/128$	$1/256$

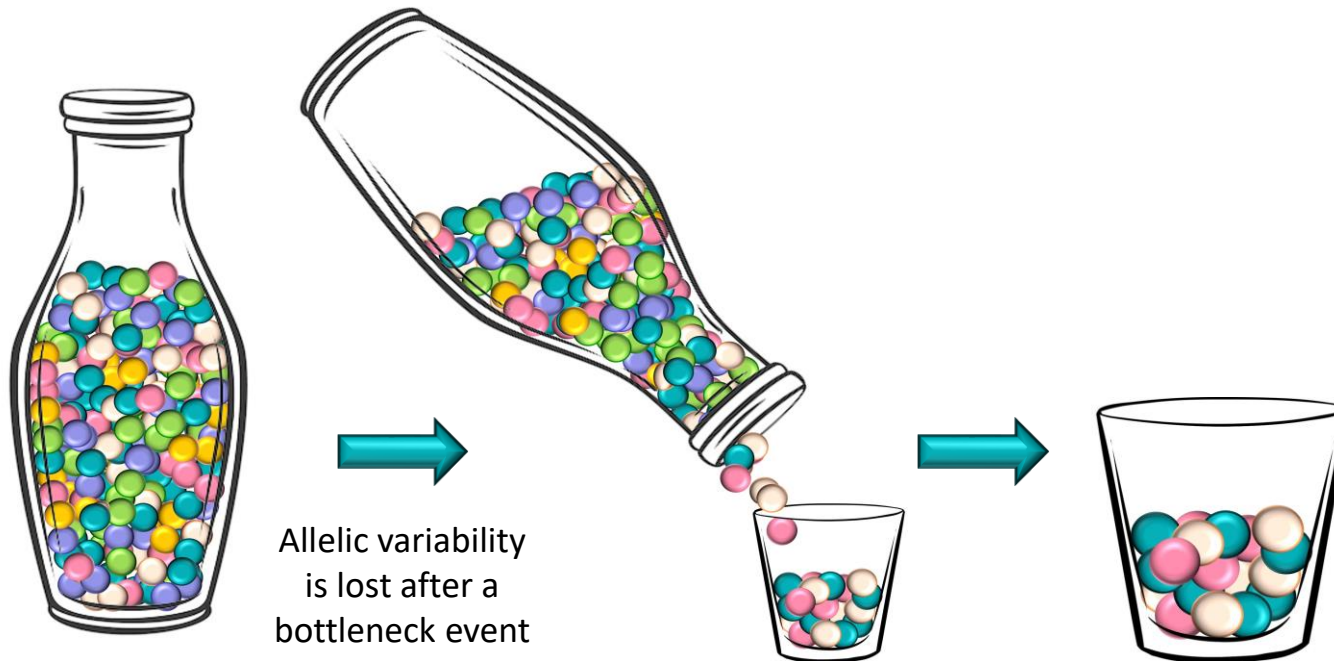
Gene Flow

- The movement of alleles from one population to another
 - Migration brings new alleles to a population
 - Allele frequencies will be changed in the target population after random mating and mixing of the alleles
 - Increasing variability helps a population survive if environment changes
- Increasing allele variability helps a population survive if environment changes drastically



Small Population Size

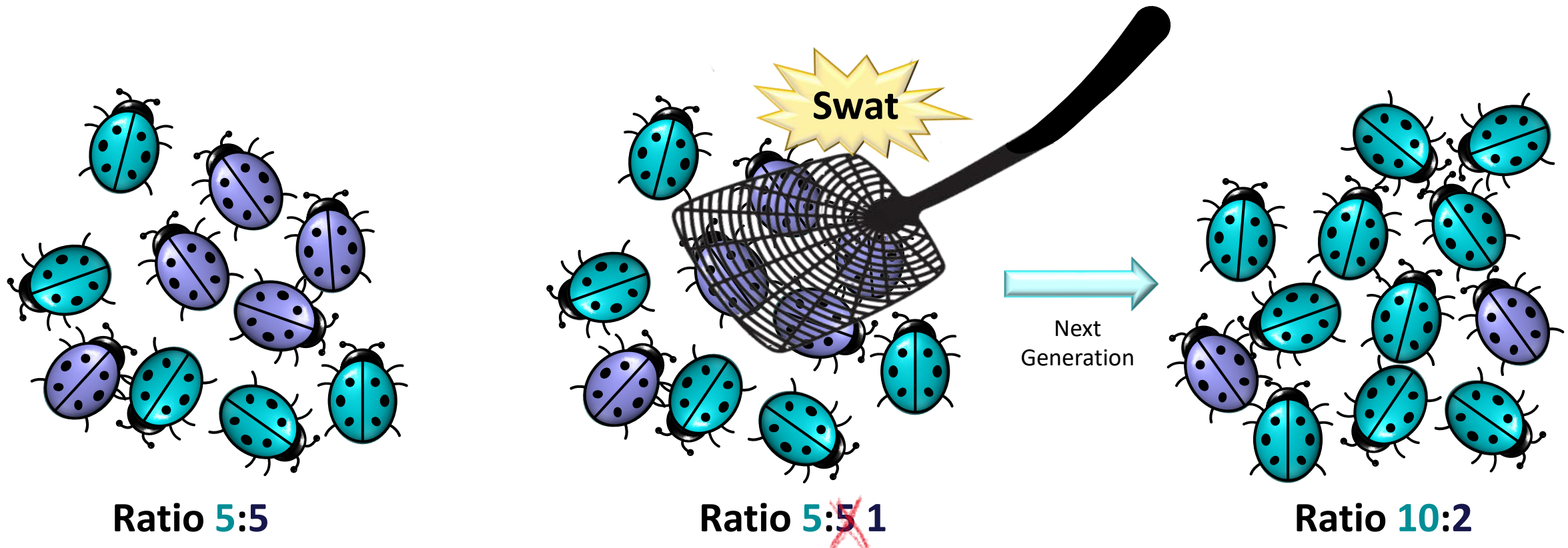
- Bottleneck to create a small population size
 - Population undergoes a severe decrease in size
 - New population reduced in genetic diversity
 - e.g. Famines, earthquakes, floods, fires, genocide



Small Population Size

- Genetic Drift in a small population

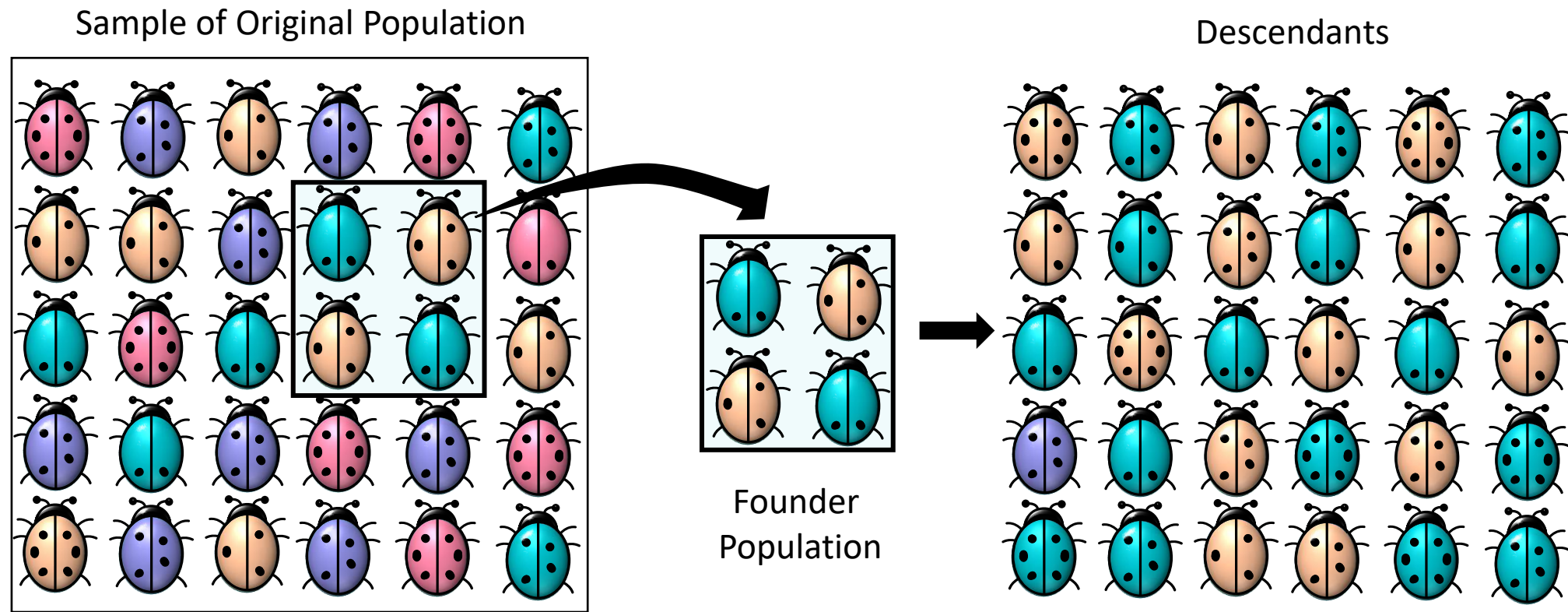
- Change in the frequency of an allelic variant due to random chance
- Gene variants may disappear, and some may fix in a small population
- Sometimes it is due to sampling error



Small Population Size

- **Founder Effect in a small population**

- Occurs when a few individuals from original population colonize an isolated region
- Loss of allelic variation
- Result may increase genetic burden of otherwise rare inherited disorders



The coat color gene of a population of mongoose is defined by a dominant allele G, which encodes a brown coat, and a recessive allele g, which encodes a grey coat. In a population, genotyping shows the distribution of alleles to be:

50 GG, 30 Gg, and 20 gg.

What is the allelic frequency of the G allele?

- A. 0.70
- B. 0.60
- C. 0.65
- D. 0.50
- E. 0.35

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$$\frac{(50 + 50 + 30)}{200}$$
$$= \frac{130}{200} = \frac{65}{100} = 0.65$$

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- | | | |
|-----------|----------------|---|
| A. 0.70 | $p + q = 1$ | $\frac{(20 + 20 + 30)}{200}$ $= \frac{70}{200} = \frac{35}{100} = 0.35$ |
| B. 0.60 | $G + g = 1$ | |
| C. 0.65 | $g = 1 - G$ | |
| D. 0.50 | $g = 1 - 0.65$ | |
| ✓ E. 0.35 | $g = 0.35$ | |

Sickle cell disease affect approximately 1/600 African Americans. The incidence is as high as 1/50 individuals in Africa. Which of the following statements best describes the mechanism leading to the high frequency of sickle cell disease in people living in Africa ?

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- B. New mutation in African population
- C. Consanguineous mating
- D. Natural selection
- E. Genetic drift

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The End

